Quantum Entanglement, Part 3

Lecture 5: Fireballs, Frames, and Four-Vectors

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Chapter 1

Fireballs, Frames, and Four-Vectors

We begin with a question left over from the discussion of high-energy collisions. A collision of two heavy nuclei creates a violently changing object, not a quiet astronomical black hole. What, then, can it mean to describe the collision by a horizon? The answer leads through a five-dimensional picture of strongly coupled matter, a set of practical questions about accelerators, and finally back to the geometry of special relativity. By the end, velocity and momentum have become four-vectors, and energy has taken its place beside the three components of momentum.

1.1 A Fireball in a Fifth Dimension

Question from the audience. The object produced in the collision is unstable and changes shape. How can such a complicated object be called a black hole?

Response. Picture a fifth spatial direction running between a floor and a ceiling. Gravity in that direction pulls excitations toward the floor. The collision deposits a hot lump of energy there; the lump spreads and changes shape, but it can still possess a horizon, temperature, and entropy.

Imagine particles, or more pictorially little strings, moving in this five-dimensional world. Excitations that fall to the floor appear in four dimensions as hadrons: protons, neutrons, mesons, and other extended composite objects. Excitations attached near the ceiling appear more elementary. The vertical location is therefore a geometric way of encoding scale: the floor represents the long-distance hadronic regime, while the ceiling represents short-distance degrees of freedom.

When two hadrons collide, two objects near the floor strike one another and make a large splash of energy. The result is not a rigid body. It is a hot horizon-bearing lump, a kind of puddle in the five-dimensional gravitational field. It is not an astronomical black hole in ordinary four-dimensional space, nor a Planck-scale black hole attached to the ceiling. It is the gravitational description of the strongly interacting fireball near the floor.

At high energy the incoming nuclei are Lorentz-contracted into thin pancakes. They collide, deposit their energy in a small region, and produce a fluid that spreads toward equilibrium. Its small viscosity was one of the surprising features of the quark–gluon plasma. In the five-dimensional description the same evolution is the relaxation of a distorted horizon.

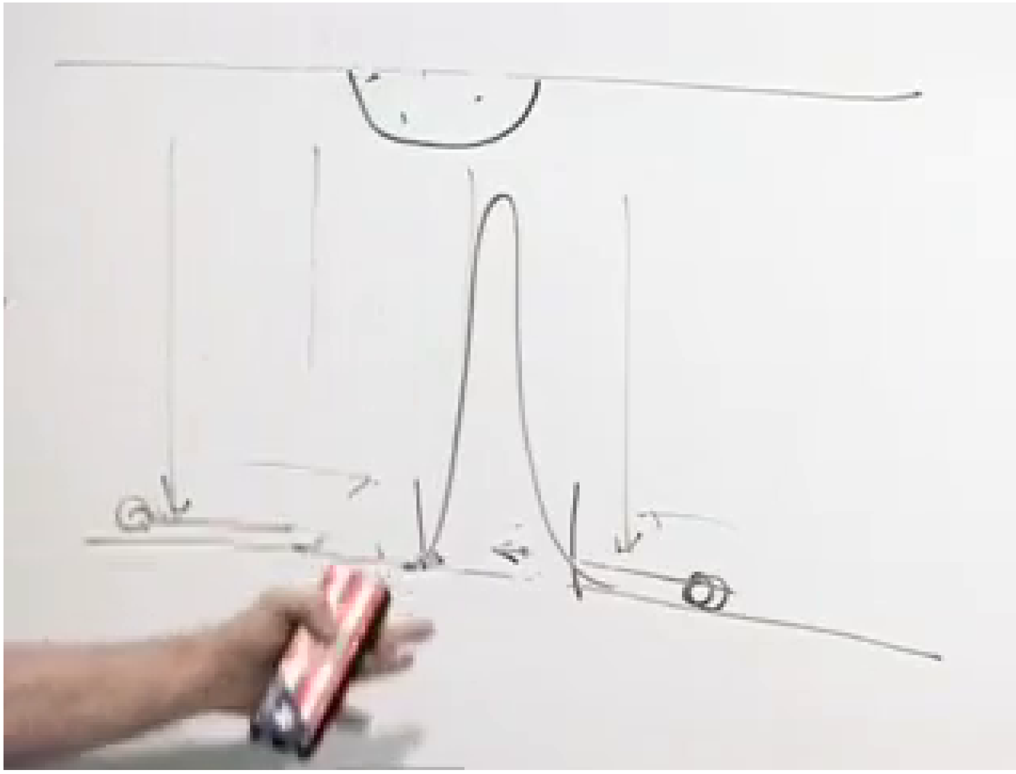

Figure 1.1: The floor-and-ceiling picture of the fifth dimension. The central horizon-bearing lump spreads along the floor after the collision.

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A horizon does not remain forever. Hawking radiation carries energy away, and in the four-dimensional description that radiation is the shower of ordinary hadrons emerging from the collision. The thousands of tracks in a detector event are, in this language, the particles boiled off the dual black hole.

1.1.1 Is the extra dimension real?

Question from the audience. Is the fifth dimension a spatial dimension, and is the line drawn between floor and ceiling literally extending in that direction?

Response. Yes. It is spacelike, and the surface drawn in that five-dimensional geometry is a horizon. Whether one calls the extra direction “real” or regards it as an equivalent mathematical description is a deeper question.

The useful statement is not that gravity replaces quantum chromodynamics. Rather, two languages may describe the same physics. In the four-dimensional language there is a hot fireball of quarks and gluons, with no ordinary gravitational field. In the five-dimensional language there is gravity and a horizon. If the correspondence is correct, an honest QCD calculation and the dual gravitational calculation must agree wherever both can be performed.

Question from the audience. QCD has neither gravity nor horizons. Is the black-hole picture adding new physics, or does it merely redescribe the fireball?

Response. In four dimensions it is a plasma of quarks and gluons; in the five-dimensional incarnation it has a gravitational description with a horizon. The advantage is calculational. Strongly coupled QCD is extremely difficult, while some horizon properties—the viscosity in particular—are comparatively easy to calculate.

The distinction between "real dimension" and "mathematical analogy" need not be settled before the description can be useful. A structure earns a physical interpretation when it joins the other variables into a simpler, more coherent set of laws. Still, one should not overstate the case: applications of gauge/gravity duality to the observed quark–gluon plasma are models of strongly coupled dynamics, not a proof that real-world QCD has a known exact gravitational dual.[1]

1.2 Why Colliders Collide Beams

The questions now become practical. Why use two moving beams instead of a single beam and a stationary target? Why are circular accelerators so large? Why are electrons and protons handled differently? All of these questions are questions about relativistic energy.

Question from the audience. Why collide two beams rather than fire one energetic beam into a stationary target?

Response. What creates new particles is the energy available in the center-of-mass frame. Two equal and opposite relativistic beams have no net momentum, so essentially all their energy is available to the collision. A fixed target forces much of the incident energy to remain as motion of the center of mass.

For two ultrarelativistic particles with opposite momenta of magnitude p, the lecture-level estimate in $c = 1$ units is

$$E_{\mathrm{cm}}^{\mathrm{collider}} \simeq 2p. \tag{1.1}$$

The corresponding fixed-target estimate grows only as the square root of the beam momentum,

$$E_{\mathrm{cm}}^{\mathrm{fixed}} \sim \sqrt{mp}. \tag{1.2}$$

The scaling, rather than the omitted numerical factor, is the point: doubling the beam energy doubles the collider energy, but it increases fixed-target center-of-mass energy only by a square root.

The invariant calculation makes the comparison precise. If P_1 and P_2 are the incoming four-momenta, then

$$s = (P_1 + P_2)^2, \qquad E_{\mathrm{cm}} = \sqrt{s}. \tag{1.3}$$

For equal counter-propagating beams of energy E, the spatial momenta cancel and $\sqrt{s} = 2E$. For a projectile of energy $E \gg m$ striking an equal-mass target at rest,

$$s = 2m^2 + 2mE, \qquad \sqrt{s} \simeq \sqrt{2mE} \qquad (c = 1). \tag{1.4}$$

This short invariant derivation supplies the factor suppressed in the spoken estimate.[2]

[1] Editorial clarification: This qualification makes explicit the modern status of the QCD application; it does not alter the five-dimensional mechanism developed here.

[2] Editorial clarification: The derivation is included to make the lecture's scaling argument reproducible.

Question from the audience. Is there a fundamental reason a circular accelerator must be so large? Could the same beam simply circulate through a smaller machine more times?

Response. Fields and materials impose limits. A particle receives a finite increase in energy per metre, so a higher-energy machine needs more effective accelerating length. A tight circle also demands greater transverse acceleration, and an accelerated charge radiates. If the ring is too small, the beam radiates energy as fast as the machine supplies it.

The bending relation is approximately

$$p = qBR, \tag{1.5}$$

so for fixed magnetic field B, a larger momentum p requires a larger radius R. The radio-frequency cavities do the work that raises the beam energy; the magnets bend and focus the beam. The broad lesson remains the one given at the board: field strength and radiative losses prevent us from compressing an arbitrarily energetic accelerator into a small ring.

Circular motion has acceleration even at constant speed because the velocity changes direction. An accelerated charge emits synchrotron radiation. A linear accelerator avoids the continuous bending loss, although the short accelerating pushes themselves are not literally radiation-free. Circular machines accept the bending cost in exchange for reusing the same apparatus on every orbit.

Question from the audience. Why were electrons associated with the straight machine at SLAC while circular high-energy colliders use protons or other hadrons?

Response. Electrons are much lighter. At a given high energy their Lorentz factor is much larger, and circular acceleration makes them radiate far more strongly. Heavier protons can circle repeatedly, gaining energy on each pass, before synchrotron losses become prohibitive.

For fixed energy and ring radius, the radiated power scales very steeply with the particle mass,

$$P_{\text{sync}} \propto \frac{E^4}{m^4 R^2}. \tag{1.6}$$

This is why the electron–proton distinction is much sharper than the phrase "the lighter particle moves faster" might suggest: both beams can already be extremely close to c, but their Lorentz factors and radiation losses are very different.[3]

Question from the audience. Why not accelerate hadrons in a straight line as well?

Response. One can. The difficulty is length. In a ring the same particle passes the accelerating structures many times; an equivalent linear hadron machine would have to be enormously long. The useful compromise is a large ring, large enough that proton radiation remains tolerable.

These accelerator questions expose exactly why Newtonian rules are insufficient. We now return to special relativity and build the language that makes such comparisons systematic.

1.3 The Garage and the Limousine

Consider a garage about twelve feet long and a stretch limousine about twenty feet long. The driver accelerates until the limousine is Lorentz-contracted, hoping that it will fit. In the garage frame the

[3]Editorial clarification: The bending relation and synchrotron scaling make the lecture's qualitative accelerator argument precise; they were not worked out on the board.

moving limousine is shorter, so it appears possible. In the limousine frame, however, the moving garage is shorter, so it appears impossible. Can the whole limousine be inside the garage?

Open both doors so that no collision with a wall distracts us. To say that the limousine is inside means that its front and rear are inside *simultaneously*. That final word contains the entire puzzle.

Draw a spacetime diagram with t vertical and x horizontal. Light rays form the 45° light cone in $c = 1$ units. The garage is at rest, so the worldlines of its two doors are vertical. The front and rear of the limousine trace two parallel sloping worldlines. The moving frame's t'-axis is tilted toward the light cone, and its lines of simultaneity are parallel to the tilted x'-axis.

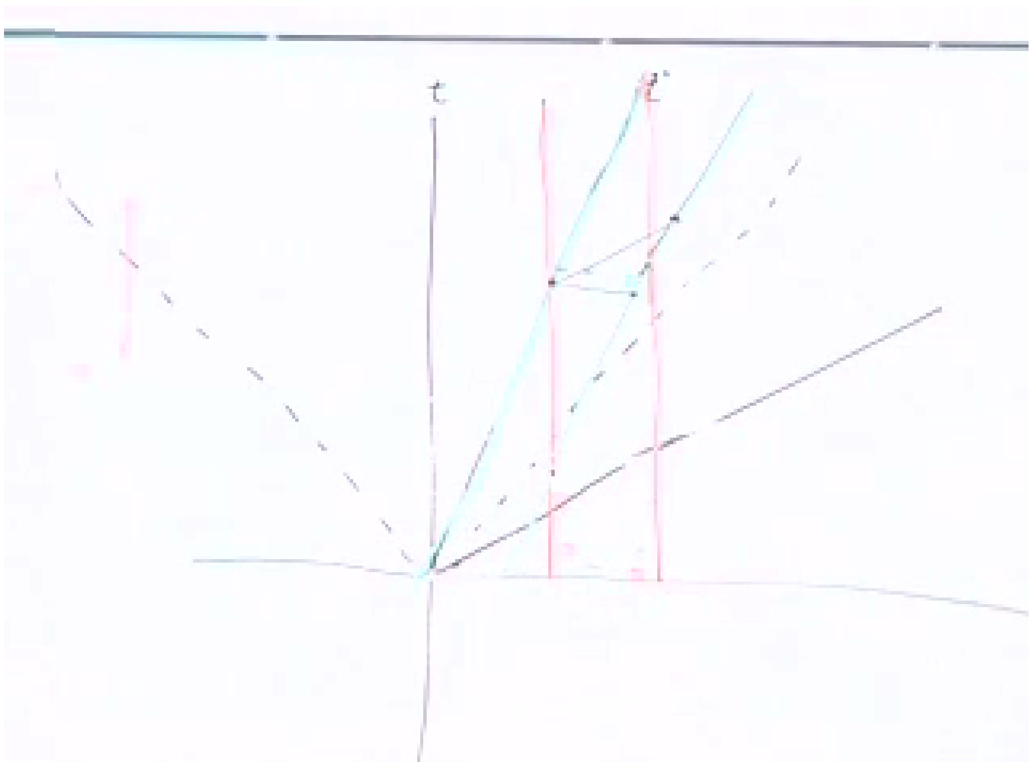

Figure 1.2: The completed garage–limousine spacetime diagram. Vertical lines bound the garage; sloping lines trace the two ends of the limousine. Horizontal and tilted cuts represent the two frames' different simultaneity slices.

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In the garage frame, a horizontal line can cross both limousine worldlines between the two garage worldlines. At that garage time the rear has entered and the front has not yet left. The contracted limousine is wholly inside.

In the limousine frame, the relevant simultaneous events lie on a tilted line. Draw such a line through the event at which the rear enters. It meets the front worldline after the front has already passed the far door. The limousine is never wholly inside according to its own simultaneity. Both statements are true because they compare different pairs of events.

Question from the audience. Do the two green worldlines and the horizontal segment between them form a parallelogram? Is that black segment the length of the vehicle?

Response. The horizontal segment is the Lorentz-contracted length, because its endpoints are simultaneous in the garage frame. The proper length is obtained from endpoints simultaneous in the limousine frame, on a line parallel to x'. The two measurements use different cuts through the same pair of worldlines.

If L_0 is the proper length, the horizontal garage-frame cut has length

$$L = \frac{L_0}{\gamma}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{1.7}$$

The formula alone does not resolve the paradox. The diagram does, because it shows which events were compared. Whenever a relativity puzzle seems to produce two incompatible answers, look for an unspoken use of "simultaneously."

1.4 Four Coordinates and Two Index Positions

To move from puzzles to laws of motion, we need notation that remembers the geometry. Package the coordinates as

$$(X^1, X^2, X^3, X^0) = (x, y, z, t), \qquad X^\mu = (x, y, z, t), \tag{1.8}$$

where μ runs through $1, 2, 3, 0$. The ordering puts time last on the page even though its index is 0, matching the convention used at the board.

Question from the audience. Are we setting $c = 1$? If ordinary units are restored, should the time coordinate be ct?

Response. Yes. We set $c = 1$ while developing the geometry and put the factors of c back when dimensions or the nonrelativistic limit matter. With conventional length units, the time coordinate is ct.

The same vector may be written with a lower index. With the sign convention used here,

$$X_\mu = (-x, -y, -z, t). \tag{1.9}$$

Thus lowering an index reverses the signs of the three spatial components and leaves the time component unchanged. For example,

$$X^\mu = (3, 2, 4, 10) \quad \Longrightarrow \quad X_\mu = (-3, -2, -4, 10). \tag{1.10}$$

These are not two spacetime points; they are two component forms of one geometric object.

Question from the audience. Which components are covariant and contravariant, and are the labels meant to be superscripts or subscripts?

Response. Upper-index components are contravariant; lower-index components are covariant. A useful memory aid from the room is "co is low, contra is high." Physicists often simply say upstairs and downstairs.

The notation earns its keep in the invariant contraction

$$X_\mu X^\mu = t^2 - x^2 - y^2 - z^2. \tag{1.11}$$

In differential form this is

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2 \qquad (c = 1). \tag{1.12}$$

For timelike motion, $d\tau$ is the time recorded by a clock moving along the worldline.

Whenever the same index appears once upstairs and once downstairs, it is understood to be summed. Einstein's convention turns

$$\sum_{\mu} X_{\mu} X^{\mu} \tag{1.13}$$

into the shorter expression in Eq. (1.11). This is not merely typographical economy. It prevents the alternating signs of Minkowski geometry from being rewritten by hand in every equation.

Question from the audience. Is τ the standard symbol for proper time?

Response. Yes. The symbol s is also sometimes used, but τ is a standard choice for proper time.

1.5 Rotations as the Easy Symmetry

A Lorentz boost mixes space and time. Before studying that matrix, consider the more familiar operation of rotating spatial axes. If the relative motion points in an arbitrary direction, we can rotate the axes until that direction becomes the new x-axis and then apply the known boost. Rotations are therefore part of the construction, not an ornamental detour.

For a rotation through angle θ about the z-axis,

$$\begin{aligned} x' &= x\cos\theta + y\sin\theta, \\ y' &= -x\sin\theta + y\cos\theta, \\ z' &= z. \end{aligned} \tag{1.14}$$

At $\theta = 0$ these equations return $x' = x$ and $y' = y$; at $\theta = \pi/2$ they exchange the axes with the expected signs.

The transformation preserves Euclidean distance. Writing $c_\theta = \cos\theta$ and $s_\theta = \sin\theta$,

$$\begin{aligned} x'^2 + y'^2 &= (xc_\theta + ys_\theta)^2 + (-xs_\theta + yc_\theta)^2 \\ &= x^2(c_\theta^2 + s_\theta^2) + y^2(s_\theta^2 + c_\theta^2) \\ &\quad + 2xyc_\theta s_\theta - 2xyc_\theta s_\theta \\ &= x^2 + y^2. \end{aligned} \tag{1.15}$$

The cross terms cancel and $c_\theta^2 + s_\theta^2 = 1$. Since $z' = z$, the full length $x^2 + y^2 + z^2$ is invariant.

The same equations become one matrix equation:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{M} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \tag{1.16}$$

In index notation,

$$x'^i = M^i{}_j x^j, \qquad i, j = 1, 2, 3. \tag{1.17}$$

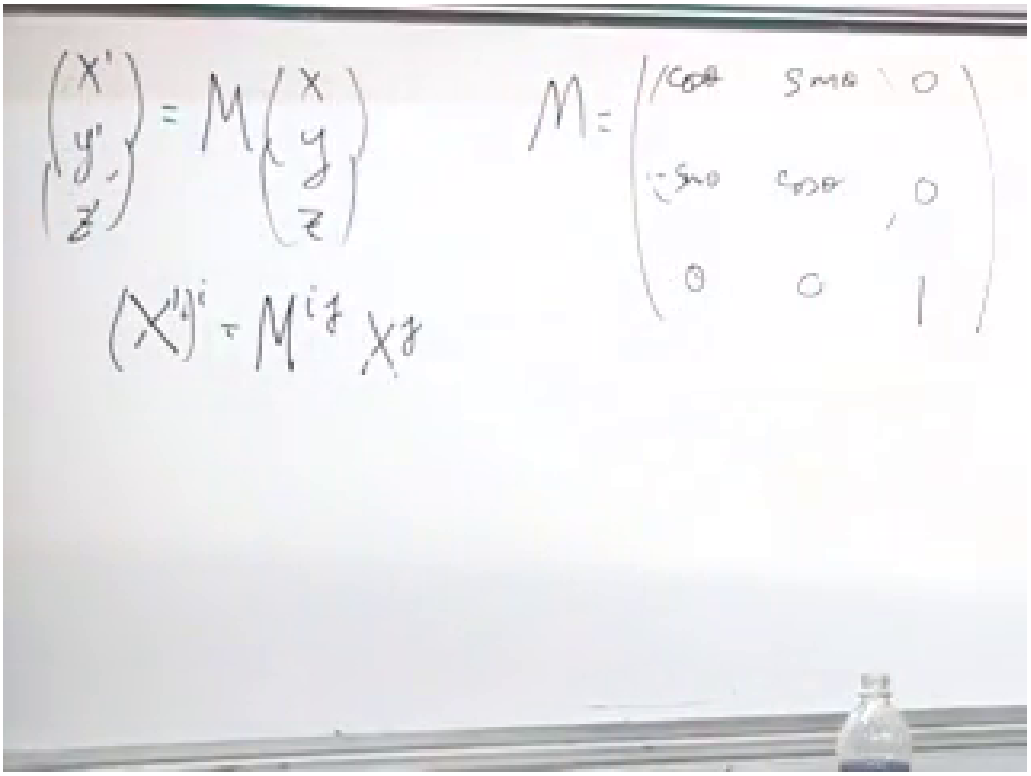

Figure 1.3: The spatial rotation written both as a matrix equation and in index notation.
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1.6 Lorentz Matrices and Composed Velocities

For motion along x, the Lorentz transformation in $c = 1$ units is

$$\begin{aligned} x' &= \gamma_v(x - vt), & y' &= y, \\ z' &= z, & t' &= \gamma_v(t - vx), \end{aligned} \tag{1.18}$$

where

$$\gamma_v = \frac{1}{\sqrt{1 - v^2}}. \tag{1.19}$$

With the component order (x, y, z, t), this becomes

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma_v & 0 & 0 & -\gamma_v v \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma_v v & 0 & 0 & \gamma_v \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{1.20}$$

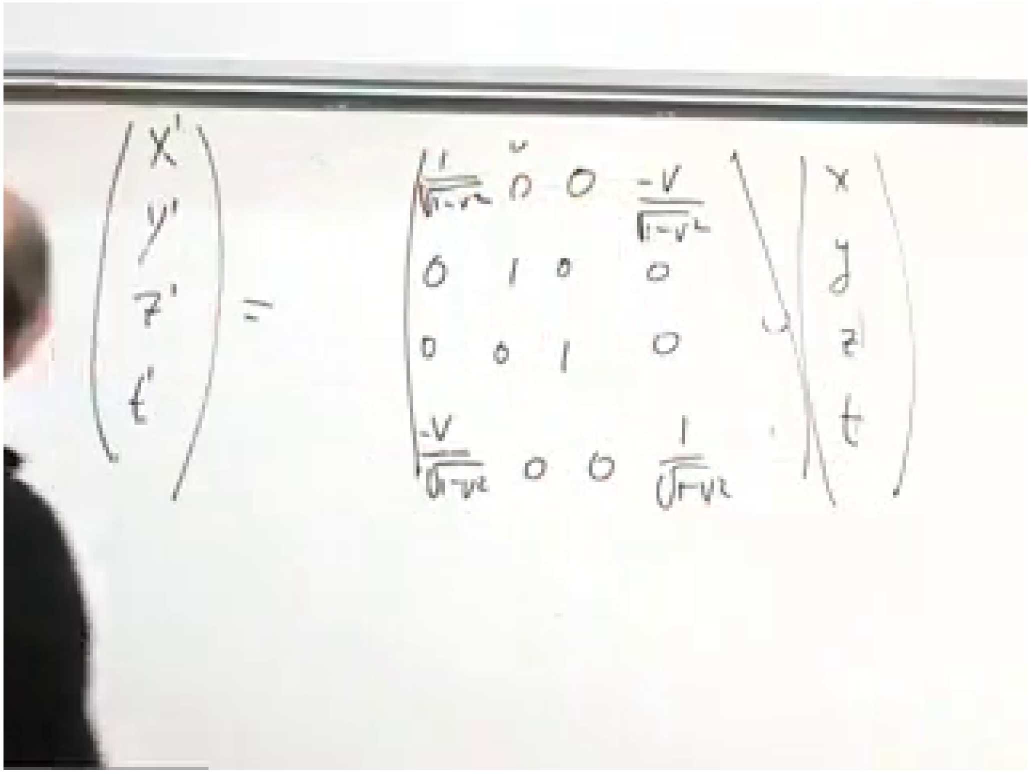

Figure 1.4: The 4×4 Lorentz matrix. Only the x and t components mix; y and z remain unchanged.
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Matrix multiplication now answers a physical question. Suppose one observer moves with velocity v relative to us and a third moves with velocity u relative to that observer, all along the same axis. Newtonian kinematics would give $w = u + v$, which would allow two velocities of $3c/4$ to combine into $3c/2$. Relativity must give a different rule.

The active part of the boost is the 2×2 matrix

$$B(v) = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}. \tag{1.21}$$

Composing the two boosts gives

$$\begin{aligned} B(u)B(v) &= \gamma_u\gamma_v \begin{pmatrix} 1 & -u \\ -u & 1 \end{pmatrix} \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \\ &= \gamma_u\gamma_v \begin{pmatrix} 1+uv & -(u+v) \\ -(u+v) & 1+uv \end{pmatrix}. \end{aligned} \tag{1.22}$$

The ratio of an off-diagonal entry to a diagonal entry identifies the velocity of the resulting boost:

$$w = \frac{u+v}{1+uv} \qquad (c = 1). \tag{1.23}$$

Restoring dimensions,

$$w = \frac{u+v}{1+uv/c^2}. \tag{1.24}$$

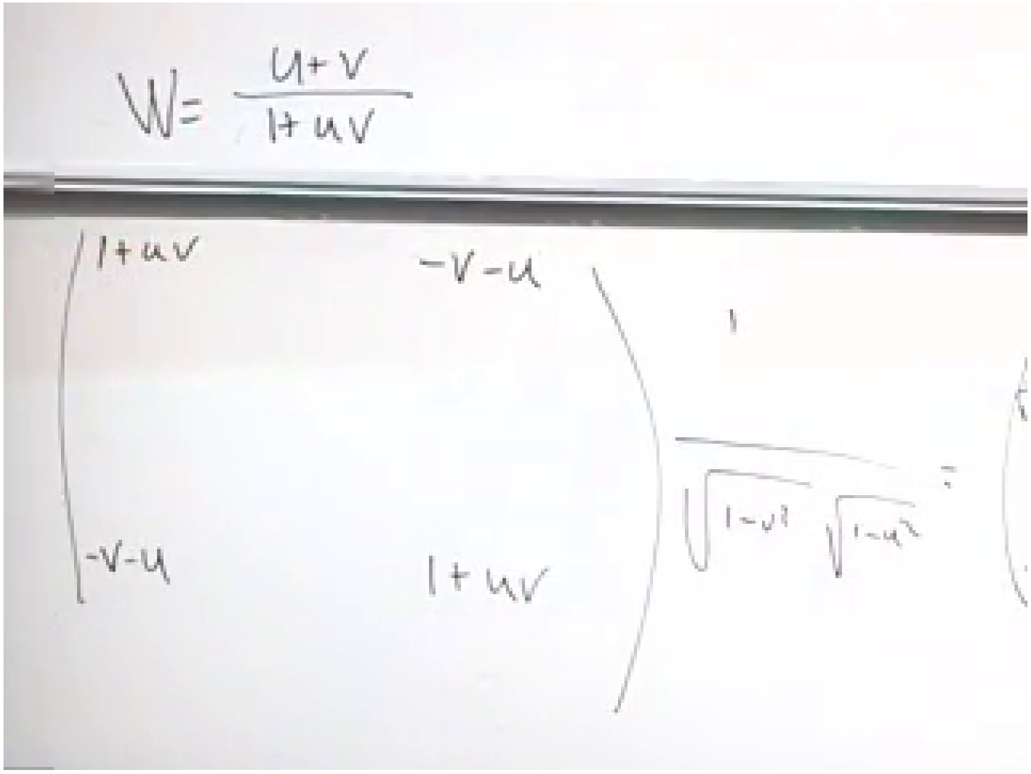

Figure 1.5: The composed-boost matrix and the velocity $w = (u+v)/(1+uv)$ in natural units.
Lecture frame: 01:15:35.

If u and v are each close to c, the numerator approaches $2c$ while the denominator approaches 2, so w approaches c, not $2c$. More generally, Eq. (1.24) maps subluminal inputs to a subluminal result. Matrix multiplication has encoded the speed limit automatically.

1.7 Four-Vectors and the Failure of Ordinary Velocity

A four-vector is any object whose four components transform under rotations and Lorentz boosts in the same way as the spacetime coordinates. If V^μ is a four-vector, then with our sign convention

$$V_\mu = (-V^1, -V^2, -V^3, V^0), \tag{1.25}$$

and its invariant norm is

$$V_\mu V^\mu = (V^0)^2 - (V^1)^2 - (V^2)^2 - (V^3)^2. \tag{1.26}$$

For two four-vectors V^μ and W^μ, the contraction

$$W_\mu V^\mu = W^0V^0 - W^1V^1 - W^2V^2 - W^3V^3 \tag{1.27}$$

is likewise invariant. The components change between frames; the contraction does not.

Ordinary velocity has only three components,

$$\mathbf{v} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right). \tag{1.28}$$

Appending $dt/dt = 1$ does not make it a useful four-vector. The fourth entry would be trivial, and the resulting object would not transform linearly under Lorentz transformations. Coordinate time t changes from one frame to another, so differentiating by t spoils the desired transformation law.

Proper time provides the cure. Along a worldline define

$$u^\mu = \frac{dX^\mu}{d\tau}. \tag{1.29}$$

The differential dX^μ is a four-vector and $d\tau$ is invariant, so their ratio is a four-vector. The adjective "proper" refers to the clock in the denominator.

The chain rule relates this object to ordinary velocity:

$$\frac{dx}{d\tau} = \frac{dx}{dt}\frac{dt}{d\tau}, \tag{1.30}$$

and similarly for y and z. From Eq. (1.12),

$$\begin{aligned}\left(\frac{d\tau}{dt}\right)^2 &= 1 - \left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2 - \left(\frac{dz}{dt}\right)^2 \\ &= 1 - v^2 \qquad (c = 1),\end{aligned} \tag{1.31}$$

and therefore

$$\frac{dt}{d\tau} = \gamma_v = \frac{1}{\sqrt{1 - v^2}}. \tag{1.32}$$

The four components are

$$u^i = \gamma_v v^i, \qquad u^0 = \gamma_v \qquad (c = 1). \tag{1.33}$$

At low speed, $u^i \simeq v^i$ and $u^0 \simeq 1$. Near the speed of light, γ_v grows without bound and proper velocity differs sharply from ordinary velocity.

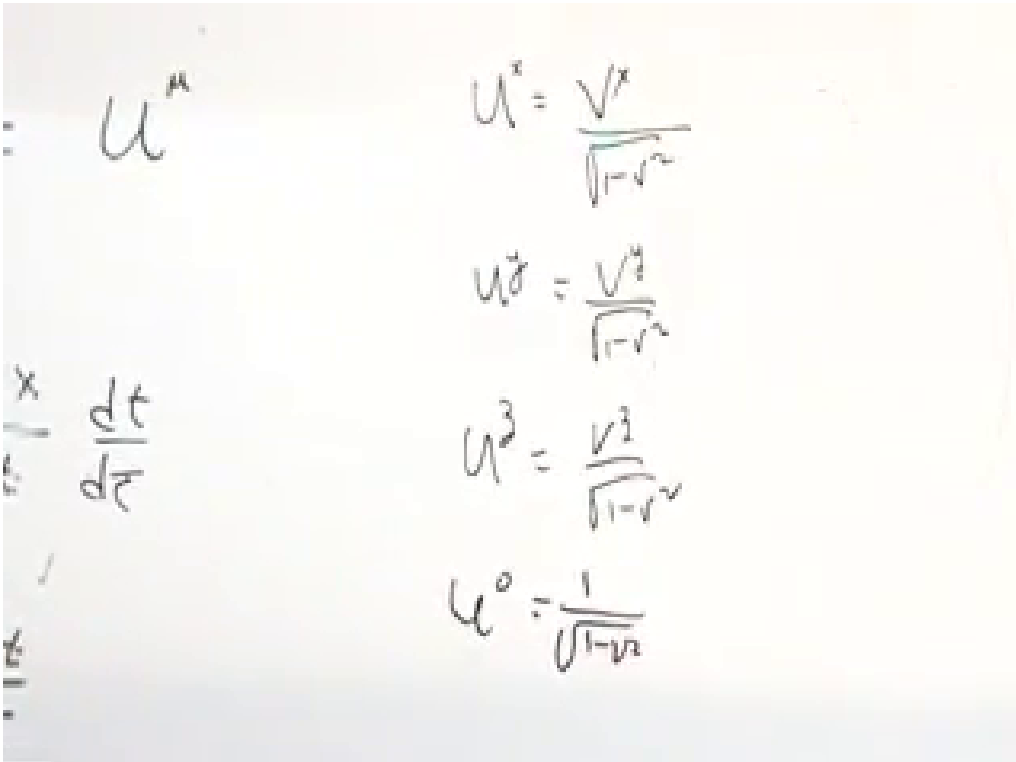

Figure 1.6: The four components of proper velocity expressed through ordinary velocity and the Lorentz factor.

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Restoring conventional units and using $X^0 = ct$ gives the standard form

$$U^\mu = (\gamma_v \mathbf{v}, \gamma_v c) \tag{1.34}$$

in the spatial-first ordering used here. Its invariant norm is

$$U_\mu U^\mu = c^2. \tag{1.35}$$

Thus every massive particle has a four-velocity of the same invariant magnitude even though its spatial velocity depends on the frame.

1.8 The Tangent to a Worldline

There is a familiar Euclidean model for Eq. (1.29). Along a curve in ordinary three-dimensional space,

$$ds^2 = dx^2 + dy^2 + dz^2. \tag{1.36}$$

The differential displacement (dx, dy, dz) points along the curve but is not a unit vector. Dividing by arc length gives

$$\mathbf{T} = \left(\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds}\right), \qquad \mathbf{T} \cdot \mathbf{T} = 1. \tag{1.37}$$

The relativistic construction is the same idea with proper time in place of Euclidean arc length. The vector $u^\mu = dX^\mu / d\tau$ is tangent to the worldline.

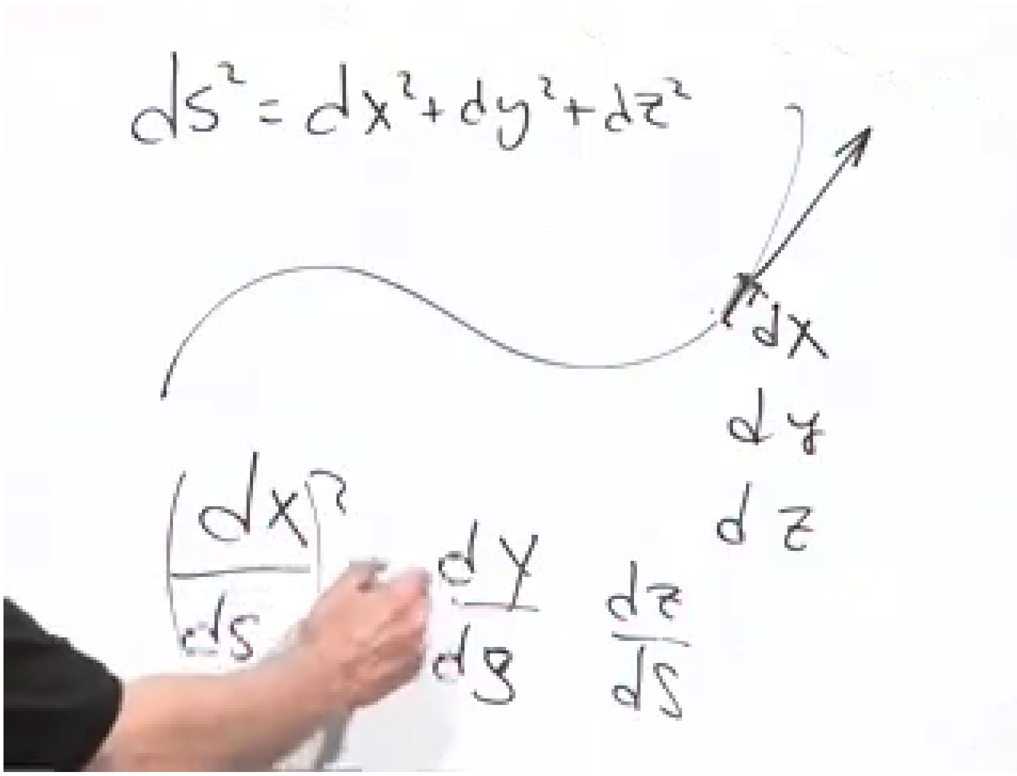

Figure 1.7: The Euclidean tangent-vector construction: divide the differential displacement by the arc-length element ds. Proper time plays the analogous role along a timelike worldline.

Lecture frame: 01:39:30.

Question from the audience. If $c = 1$, are proper time and proper distance the same?

Response. Setting $c = 1$ puts time and distance in the same units; it does not erase the minus signs of spacetime geometry. In the Euclidean analogy, s is ordinary distance measured along a spatial curve. Along a timelike worldline, τ is the elapsed time measured by a clock. The constructions are analogous, but the geometries are different.

Differentiating once more defines proper acceleration,

$$a^\mu = \frac{du^\mu}{d\tau}. \tag{1.38}$$

This is the quantity that must replace $d^2\mathbf{x}/dt^2$ when Newton's second law is made relativistic. Since $u_\mu u^\mu$ is constant,

$$\frac{d}{d\tau}(u_\mu u^\mu) = 2u_\mu a^\mu = 0, \tag{1.39}$$

so four-acceleration is Minkowski-orthogonal to four-velocity.[4]

[4]Editorial clarification: The orthogonality follows immediately from the invariant norm and is included as the next mathematical consequence of the definition.

1.9 Four-Momentum and Energy

Momentum is our final example because it is conserved in collisions and decays. The particle has a fixed intrinsic mass m, often called its rest mass. It does not vary with velocity. Relativistic momentum is that fixed mass times four-velocity:

$$p^\mu = mu^\mu. \tag{1.40}$$

In $c = 1$ units,

$$p^i = \gamma_v m v^i, \qquad p^0 = \gamma_v m. \tag{1.41}$$

At small velocity $\gamma_v \simeq 1$, so the spatial components reduce to the Newtonian momentum $m\mathbf{v}$. At high velocity they grow more rapidly than $m\mathbf{v}$.

The time component is energy. With conventional units and $X^0 = ct$,

$$P^\mu = (\mathbf{p}, E/c) = (\gamma_v m\mathbf{v}, \gamma_v mc), \tag{1.42}$$

and hence

$$E = \gamma_v mc^2 = mc^2\left(1 - \frac{v^2}{c^2}\right)^{-1/2}. \tag{1.43}$$

To recover ordinary mechanics, let $\epsilon = v^2/c^2 \ll 1$. The binomial expansion gives

$$(1-\epsilon)^p = 1 - p\epsilon + O(\epsilon^2). \tag{1.44}$$

Taking $p = -1/2$,

$$\begin{aligned} E &= mc^2\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \\ &= mc^2\left[1 + \frac{1}{2}\frac{v^2}{c^2} + O\left(\frac{v^4}{c^4}\right)\right] \\ &= mc^2 + \frac{1}{2}mv^2 + O\left(\frac{mv^4}{c^2}\right). \end{aligned} \tag{1.45}$$

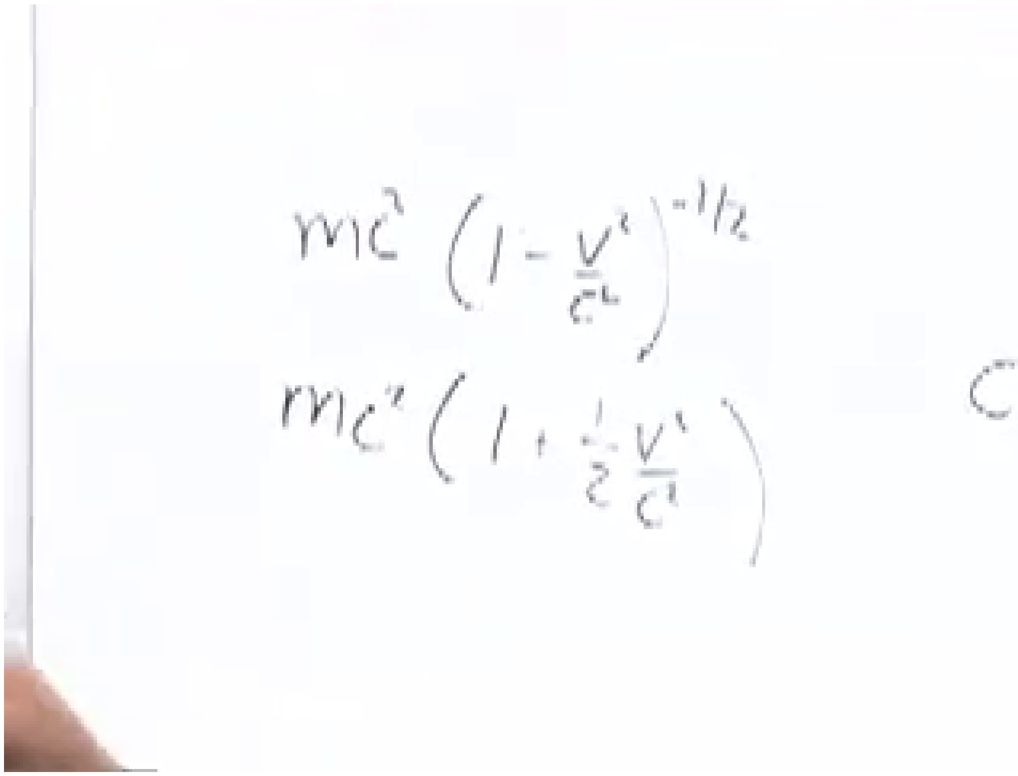

Figure 1.8: The low-velocity expansion of relativistic energy, revealing rest energy and Newtonian kinetic energy.

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The first term, mc^2, is present even at rest. In a Newtonian problem with a fixed collection of particles and unchanging masses, that term adds the same constant before and after the motion, so only $\frac{1}{2}mv^2$ affects the familiar bookkeeping. Relativity reveals that the discarded constant was physical energy all along.

Energy and momentum are therefore not unrelated conservation laws. They are the four components of one conserved vector. A Lorentz boost mixes them: what one observer calls energy contributes partly to another observer's momentum, and conversely. Nevertheless, for an isolated system the total four-momentum is the same before and after every collision or decay in every inertial frame.

This is the threshold of relativistic mechanics. The next tasks are to express force through proper acceleration and to understand relativistic wave equations, especially the equations governing electromagnetism and light.